Mini Project II

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Result

	Sample Size			
	50	100	200	400
Case 1	0.260	0.426	0.708	0.948
Case 2	0.088	0.105	0.190	0.300
Case 3	0.166	0.320	0.544	0.863

Figure 1

Discussion

Based on this simulation study, we find that large sample sizes positevly impact power assuming equal sample sizes and adminstrative sampling. Additionally, between cases, cases with greater effect sizes (i.e., larger differences in hazard rates between intervention and control groups) have greater power. For instance, Case 1 - the largest effect size - has a larger power across all sample sizes in comparison to Case 2 or Case 3.

For the study at hand, we recommend comparing the intervention with a constant hazard rate of 0.1 with the control with a sample size of 400 patients in both groups for the most robust results. With these specifications, we will best detect true differences in days to readmission among patients with heart failue between the intervention and control groups. Using a hazard rate of 0.16 for the first week and then a subsequent rate of 0.04 with the same sample size (n=400) is also a viable option for assessing intervention effects based on power analysis.

Appendix: R scripts

```
library(nphsim)
library(knitr)
library(kableExtra)
```

```r
# sample sizes
n.sim = c(50, 100, 200, 400)

# probablities for intervention
probs = list(c(0.1, 0.1), c(0.25, 0.025), c(0.16, 0.04))

# final rejection count matrix
rej = matrix(0, nrow = 4, ncol = 3)

# simulation with 1000 evals
for (i in (1:1000)){
  # loop through each sample size
  for (j in seq_along(n.sim)){

    # hazard rate for control
    x = rpwexp(n.sim[j], rate = c(0.2, 0.05, 0.05), intervals = c(7, 23))

    # censored values for control
    sx <- ifelse(x > 30, 0, 1)

    # control group df
    ctr_df <- data.frame(
      time = x,
      event = sx,
      trt = 0
    )
    # loop through each probability scenerio
    for (z in seq_along(probs)){
      prob = probs[z]

      # hazard rate for intervention
      x_intv = rpwexp(n.sim[j],
                      rate = c(prob[[1]][1],
                               prob[[1]][2],
                               prob[[1]][2]),
                      intervals = c(7,23))

      # censored values for intervention
      sx_intv = ifelse(x_intv > 30, 0, 1)

      # intervention group df
      intv_df <- data.frame(
        time = x_intv,
        event = sx_intv,
        trt = 1
      )

      # combine intervention and control dfs
      df = rbind(ctr_df, intv_df)

      # get p-value from log-rank test
      p = survdiff(Surv(time, event)~trt, data=df)$p
```

```
      # add rejection count to matrix if p-value < 0.05
      if (p < 0.05) {
        rej[j, z]= rej[j, z] + 1
      }
    }
  }
}

# divide rejection count by simulation length (1000) to get power
pwr = t(rej/1000)
rownames(pwr) = c("Case 1", "Case 2", "Case 3")
colnames(pwr) = c(50, 100, 200, 400)
```

```
kable(pwr, booktable = TRUE,
      caption = "Power of Varying Sample Sizes by Case") %>%
  add_header_above(header = c("Sample Size" = 5)) %>%
  kable_classic(full_width = F, html_font = "Cambria")
```

Power of Varying Sample Sizes by Case

	Sample Size			
	50	100	200	400
Case 1	0.267	0.454	0.719	0.934
Case 2	0.088	0.118	0.172	0.336
Case 3	0.166	0.321	0.580	0.848